

## Abstract

This note aims to introduce two interesting concepts of Matroid Theory, namely (i). Matroid Intersection: What can we say about the largest size of a common independent set of two matroids on the same ground set? and (ii). Matroid Union: Given a family of matroids on the same ground set, can we define a new matroid on that set that looks like the "union" of those ones? For each concept, we will give a theorem and an application in Graph Theory. Finally, we will show that the two theorems are actually equivalent.

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## 1 Matroid Intersection

We begin this section with a motivating example:

**Example 1.0.1.** Let  $G = (V_1, V_2, E)$  be a bipartite graph. For each vertex  $v \in V_1 \sqcup V_2$ , let  $\delta(v)$  denote the set of edges incident with  $v$ , then we may define  $\mathcal{M}_v$  to be the rank one (uniform) matroid on  $\delta(v)$ . Now, define:

$$\mathcal{M}_1 := \bigoplus_{v \in V_1} \mathcal{M}_v \quad \mathcal{M}_2 := \bigoplus_{v \in V_2} \mathcal{M}_v$$

so that  $\mathcal{M}_1$  and  $\mathcal{M}_2$  are matroid on the same ground set  $E$ . Note that a subset, say  $S \subseteq E$ , is independent in  $\mathcal{M}_i$  if and only if  $|S \cap \delta(v)| \leq 1$  for every  $v \in V_i$ . Therefore,  $S$  is independent in both  $\mathcal{M}_1$  and  $\mathcal{M}_2$  if and only if  $S$  is a matching. In particular, finding a maximum size matching in our bipartite graph is equivalent to finding a maximum size subset of  $E$  that is independent in both  $\mathcal{M}_1$  and  $\mathcal{M}_2$ . Now, how to do this?

**Theorem 1.0.2** (Matroid Intersection). *Let  $\mathcal{M}$  and  $\mathcal{N}$  be matroids on the same ground set  $E$ . Then*

$$\max_{I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})} |I| = \min_{X \subseteq E} \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X)$$

One direction is easy to see, and the other is hard. We postpone the proof of the hard direction.

*Proof of the easy direction.* We show that LHS  $\leq$  RHS. To see this, choose  $I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$  s.t.  $|I|$  is maximum. Then, for any  $X \subseteq E$ , since  $I$  is independent in  $\mathcal{M}$ , we have  $|I \cap X| \leq \text{rk}_{\mathcal{M}}(X)$ ; and since  $I$  is independent in  $\mathcal{N}$ , we have  $|I \cap X^c| \leq \text{rk}_{\mathcal{N}}(X^c) = \text{rk}_{\mathcal{N}}(E \setminus X)$ . Therefore

$$|I| = |I \cap X| + |I \cap X^c| \leq \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X)$$

This holds for any subset  $X \subseteq E$ , taking the minimum yields the desired result. □

**Corollary 1.0.3** (König). *Let  $G$  be an arbitrary bipartite graph. Then the size of the maximum matching in  $G$  is equal to the size of the minimum vertex cover of  $G$ .*

*Proof.* Recall that a vertex cover of  $G$  is a subset of  $V(G)$ , say  $T$ , s.t. each edge in  $E(G)$  is incident with at least one vertex in  $T$ . By our discussion above, the size of the maximum matching in  $G$  is the size of a maximum independent set in both matroids. Let  $S \subseteq E$  be a such matching, by Theorem 1.0.2, we may choose a subset  $X \subseteq E$  s.t.

$$|S| = \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X)$$

Let  $U_1(U_2, \text{resp})$  be the vertices in  $V_1(V_2, \text{resp})$  incident with  $X(E \setminus X, \text{resp})$ . Then  $\text{rk}_{\mathcal{M}}(X) = |U_1|$ ,  $\text{rk}_{\mathcal{N}}(E \setminus X) = |U_2|$ . Note that  $U_1 \sqcup U_2$  is a vertex cover. Hence

$$\text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X) = |U_1| + |U_2| = \text{the size of a vertex cover}$$

as desired.  $\square$

The following is another application of Theorem 1.0.2: For any set  $E$ , we say a **coloring** of  $E$  is a function  $f$  with domain  $E$ . We say a subset  $X \subseteq E$  is a **rainbow** of  $f$  if  $f|_X$  is injective.

**Corollary 1.0.4.** *Let  $V$  be a  $\mathbb{K}$ -vector space with  $\dim V = d$ . Let  $E \subseteq V$  be a finite subset and let  $f$  be a coloring of  $E$ . Then exactly one of the following happens:*

1. *There is a rainbow basis  $B \subseteq E$*
2. *There exists  $X \subseteq E$  so that  $\text{rk}(X) + |f(E \setminus X)| < d$*

Before we prove this corollary, we first state an equivalent version of Theorem 1.0.2:

**Theorem 1.0.5** (Matroid Intersection, Version 2). *Let  $\mathcal{M}$  and  $\mathcal{N}$  be matroids on the same ground set  $E$ . Let  $k$  be an arbitrary non-negative number. Then exactly one of the following happens:*

1. *There is a common independent set of size  $k$*
2. *There is a subset  $X \subseteq E$  s.t.  $\text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X) < k$*

*Proof of Corollary 1.0.4.* Our job is to specify the two correct matroids. Let  $\mathcal{M}$  be the vector matroid on  $E$ . Let  $C := \text{im } f$  (i.e., the colors of the coloring  $f$ ). For each  $c \in C$ , define a rank-one uniform matroid on  $f^{-1}(c)$ , denoted  $\mathcal{N}_c$ . Then, let

$$\mathcal{N} := \bigoplus_{c \in C} \mathcal{N}_c$$

Note that a rainbow basis is exactly a common independent set of size  $d$ ; and  $\text{rk}_{\mathcal{N}}(X)$  is exactly  $|f(X)|$ , for all  $X \subseteq E$ .  $\square$

To close this section, we finally give the proof of the matroid intersection theorem.

*Proof of the theorem 1.0.2.* We prove the equivalent version 1.0.5 by induction on  $|E|$  with the base case of the empty matroid holding trivially. Choose  $e \in E$  arbitrarily.

**Deletion:** Suppose we delete  $e$  and consider  $\mathcal{M} \setminus e$  and  $\mathcal{N} \setminus e$  on  $E := E \setminus e$ . If  $\exists I \in \mathcal{I}(\mathcal{M} \setminus e) \cap \mathcal{I}(\mathcal{N} \setminus e)$  with  $|I| = k$ , then  $I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$  and we are done. Otherwise, by induction hypothesis we may choose  $A \subseteq E'$  s.t.  $\text{rk}_{\mathcal{M} \setminus e}(A) + \text{rk}_{\mathcal{N} \setminus e}(E' \setminus A) \leq k - 1$ . But note that  $\text{rk}_{\mathcal{M} \setminus e}(A) = \text{rk}_{\mathcal{M}}(A)$  and  $\text{rk}_{\mathcal{N} \setminus e}(E' \setminus A) = \text{rk}_{\mathcal{N}}(E' \setminus A)$ . Therefore, we have:

$$\text{rk}_{\mathcal{M}}(A) + \text{rk}_{\mathcal{N}}(E' \setminus A) \leq k - 1 \quad (1)$$

If  $e \in \text{span}_{\mathcal{M}}(A)$ , then  $\text{rk}_{\mathcal{M}}(A + e) = \text{rk}_{\mathcal{M}}(A)$  and hence we may choose  $X := A + e$  so that

$$\text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X) \leq k - 1$$

and we win. So we may assume that  $e \notin \text{span}_{\mathcal{M}}(A)$  and similarly  $e \notin \text{span}_{\mathcal{N}}(E' \setminus A)$ . It follows that  $e$  is not a loop in  $\mathcal{M}$  or  $\mathcal{N}$ .

**Contraction:** Now consider  $\mathcal{M}/e$  and  $\mathcal{N}/e$  on  $E'$ . If  $\exists I \in \mathcal{I}(\mathcal{M}/e) \cap \mathcal{I}(\mathcal{N}/e)$  with  $|I| = k$  with  $|I| = k - 1$ , then  $I + e \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$  since  $e$  is not a loop and we are done. Otherwise, by induction hypothesis we may choose  $B \subseteq E'$  s.t.  $\text{rk}_{\mathcal{M}/e}(B) + \text{rk}_{\mathcal{N}/e}(E' \setminus B) \leq k - 2$ . But note that  $\text{rk}_{\mathcal{M}}(B + e) = \text{rk}_{\mathcal{M}/e}(B) + 1$  and  $\text{rk}_{\mathcal{N}}(E \setminus B) = \text{rk}_{\mathcal{N}/e}(E' \setminus B) + 1$ . Therefore, we have:

$$\text{rk}_{\mathcal{M}}(B + e) + \text{rk}_{\mathcal{N}}(E \setminus B) \leq k \quad (2)$$

Combining (1) and (2) and applying submodularity yields:

$$\begin{aligned} 2k - 1 &\geq \text{rk}_{\mathcal{M}}(A) + \text{rk}_{\mathcal{M}}(B + e) + \text{rk}_{\mathcal{N}}(E' \setminus A) + \text{rk}_{\mathcal{N}}(E \setminus B) \\ &\geq \text{rk}_{\mathcal{M}}(A \cap B) + \text{rk}_{\mathcal{M}}(A \cup B + e) + \text{rk}_{\mathcal{N}}(E' \setminus (A \cup B)) + \text{rk}_{\mathcal{N}}(E \setminus (A \cap B)) \end{aligned}$$

It follows that at least one of the following holds:

**case 1.**  $\text{rk}_{\mathcal{M}}(A \cap B) + \text{rk}_{\mathcal{N}}(E \setminus (A \cap B)) \leq k - 1$

**case 2.**  $\text{rk}_{\mathcal{M}}(A \cup B + e) + \text{rk}_{\mathcal{N}}(E' \setminus (A \cup B)) \leq k - 1$

In case 1, the set  $X_1 := A \cap B$  satisfies the desired bound, and in case 2, the set  $X_2 := A \cup B + e$  works. So the proof is complete.  $\square$

## 2 Matroid Union

Let  $\mathcal{M}_1, \dots, \mathcal{M}_k$  be matroids on the same ground set  $E$ .

**Definition 2.0.1.** Let

$$\mathcal{I} := \left\{ \bigcup_{i=1}^k I_i \mid I_i \in \mathcal{I}(\mathcal{M}_i), 1 \leq i \leq k \right\}$$

Then, convince yourself that  $\mathcal{I}$  is the collection of independent sets of a matroid on  $E$ . We call this matroid the **union** of  $\mathcal{M}_1, \dots, \mathcal{M}_k$ , denoted

$$\bigvee_{i=1}^k \mathcal{M}_i$$

*Remark.* You can either directly verify the axioms of independent sets or use Rado's Theorem to construct the matroid explicitly. For the latter, see another note of mine.

In fact, we can write down the rank function of that matroid:

**Theorem 2.0.2** (Matroid Union). *Let  $\text{rk}_i$  be the rank function of each matroid  $\mathcal{M}_i$ . Let  $\mathcal{M} := \bigvee_{i=1}^k \mathcal{M}_i$ . Then*

$$\text{rk}_{\mathcal{M}}(E) = \min_{X \subseteq E} |E \setminus X| + \sum_{i=1}^k \text{rk}_i(X)$$

Although the above theorem only gives the rank of the entire ground set, we can actually deduce the rank of any subset immediately:

**Corollary 2.0.3.** *We inherit the above settings. Then, for any  $X \subseteq E$ :*

$$\text{rk}_{\mathcal{M}}(X) = \min_{Y \subseteq X} |X \setminus Y| + \sum_{i=1}^k \text{rk}_i(Y)$$

*Proof.* Apply The Matroid Union Theorem 2.0.2 on  $\mathcal{M}_1|_X, \dots, \mathcal{M}_k|_X$  where  $\mathcal{M}_i|_X$  is the matroid  $\mathcal{M}_i$  restricted on  $X$ .  $\square$

We will give two proofs of Theorem 2.0.2, both of which rely on the Matroid Intersection Theorem 1.0.2. In fact, there is a third proof using Rado's theorem, which is the easiest one. To see this, again, check another note of mine. However, before, this, let's see an direct application:

Let  $\mathcal{M}$  be a matroid on  $E$ , and  $k$  be a positive integer. Set

$$\mathcal{M}^k := \underbrace{\mathcal{M} \vee \dots \vee \mathcal{M}}_{k \text{ times}}$$

So a subset  $X \subseteq E$  is independent in  $\mathcal{M}^k$  iff it can be expressed as a (disjoint) union of  $k$  sets, each of which is independent in  $\mathcal{M}$ . For a graph  $G = (V, E)$  and a partition  $V_1 \sqcup \dots \sqcup V_t$  of  $V$  we let  $E(V_1, \dots, V_t)$  denote the set of edges  $(u, v) \in E$  so that  $u \in V_i$  and  $v \in V_j$  where  $i \neq j$ .

**Corollary 2.0.4** (Tutte, Nash-Williams). *Let  $G = (V, E)$  be a connected graph and  $k$  be a positive integer,  $n := |V|$ . Then, exactly one of the following holds:*

- (i)  $G$  has  $k$  pairwise edge-disjoint spanning trees.
- (ii) There exists a partition  $V_1 \sqcup \dots \sqcup V_t$  of  $V$  s.t.  $|E(V_1, \dots, V_t)| < k(t-1)$ .

*Proof.* First, note that the two conditions are mutually exclusive. Suppose we have  $k$  pairwise edge-disjoint spanning trees, say  $T_1, \dots, T_k$  and a partition  $V_1 \sqcup \dots \sqcup V_t$  of  $V$ . In order to connect those blocks, each  $T_i$  has to contain at least  $t-1$  edges in  $E(V_1, \dots, V_t)$ , so  $|E(V_1, \dots, V_t)| \geq k(t-1)$ .

Now, let  $\mathcal{M}$  be the usual graphic matroid on  $E$  and let  $\mathcal{N} := \mathcal{M}^k$ . Suppose that (i) does not hold; this implies  $\text{rk}_{\mathcal{N}}(E) < k(n-1)$ . Hence, by Matroid Union, we may choose  $X \subseteq E$  s.t.

$$|E \setminus X| + k \cdot \text{rk}_{\mathcal{M}}(X) < k(n-1)$$

Among all feasible  $X$ , we choose a maximal one. Let  $G_1, \dots, G_t$  be the components of the graph  $(V, X)$  and let  $V_i := V(G_i)$ . If there exists an edge  $e \in E \setminus X$  s.t.  $e$  has both ends in  $V_i$  for some  $i \in [t]$ , then  $e \in \text{span}_{\mathcal{M}}(X)$  and hence  $X + e$  contradicts our choice of  $X$ . It follows that  $E \setminus X = E(V_1, \dots, V_t)$  and  $\text{rk}_{\mathcal{M}}(X) = n - t$ . Therefore

$$|E \setminus X| + k \cdot \text{rk}_{\mathcal{M}}(X) = |E(V_1, \dots, V_t)| + k(n-t) < k(t-1)$$

So  $|E(V_1, \dots, V_t)| < k(t-1)$ . i.e., (ii) holds. □

Now, we give the first proof of the Matroid Union:

*Proof of Theorem 2.0.2.* For  $i \in [k]$  let  $\mathcal{M}'_i$  be the matroid on  $\{i\} \times E$  given by the rule that  $\mathcal{M}'_i \cong \mathcal{M}_i$ . Now, define two new matroids on  $\hat{E} := [k] \times E$  as follow:  $\mathcal{M} := \bigoplus_{i=1}^k \mathcal{M}'_i$  and  $\mathcal{N} := \bigoplus_{e \in E} \mathcal{N}_e$  where  $\mathcal{N}_e$  is the rank-1 uniform matroid on  $[k] \times \{e\}$ .

|          |                                                   |     |     |         |          |                                                    |                 |                 |         |
|----------|---------------------------------------------------|-----|-----|---------|----------|----------------------------------------------------|-----------------|-----------------|---------|
|          | $e$                                               | $f$ | $g$ | $\dots$ |          | $e$                                                | $f$             | $g$             | $\dots$ |
| 1        | $\mathcal{M}'_1$                                  |     |     |         |          | $\mathcal{N}_e$                                    | $\mathcal{N}_f$ | $\mathcal{N}_g$ | $\dots$ |
| 2        | $\mathcal{M}'_2$                                  |     |     |         | 2        |                                                    |                 |                 |         |
| $\vdots$ | $\vdots$                                          |     |     |         | $\vdots$ |                                                    |                 |                 |         |
| $k$      | $\mathcal{M}'_k$                                  |     |     |         | $k$      |                                                    |                 |                 |         |
|          | $\mathcal{M} := \bigoplus_{i=1}^k \mathcal{M}'_i$ |     |     |         |          | $\mathcal{N} := \bigoplus_{e \in E} \mathcal{N}_e$ |                 |                 |         |

Let  $X \subseteq \hat{E}$  and let  $X_i := \{e \in E : (i, e) \in X\}$ . Note that

$$X \in \mathcal{I}(\mathcal{M}) \iff X_i \in \mathcal{I}(\mathcal{M}'_i), \forall i \in [k] \quad X \in \mathcal{I}(\mathcal{N}) \iff X_1, \dots, X_k \text{ are pairwise disjoint}$$

Therefore, the maximum size of a common independent set in  $\mathcal{M}$  and  $\mathcal{N}$  equals the maximum size of an independent set in  $\bigvee_{i=1}^k \mathcal{M}_i$ , which is just the rank of the ground set  $E$ . Thanks to the Matroid Intersection Theorem 1.0.2, this further implies:

$$\text{rk}_{\bigvee_{i=1}^k \mathcal{M}_i}(E) = \max_{I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})} |I| = \min_{Y \subseteq \hat{E}} \text{rk}_{\mathcal{M}}(Y) + \text{rk}_{\mathcal{N}}(\hat{E} \setminus Y)$$

Now, among all the subsets of  $\hat{E}$  attaining the minimum above, choose one to be minimal, denoted  $Y$ . Suppose for a contradiction that  $\exists i, j \in [k]$  s.t.  $(i, e) \in Y$  and  $(j, e) \notin Y$  for some  $e \in E$ , then removing  $(i, e)$  from  $Y$  will not increase  $\text{rk}_{\mathcal{M}}(-)$  and  $\text{rk}_{\mathcal{N}}(\hat{E} \setminus -)$ , and thus contradicts the choice of  $Y$ . It follows that  $Y$  would be the form  $[k] \times Z$  for some  $Z \subseteq E$ . Therefore

$$\begin{aligned} \text{rk}_{\bigvee_{i=1}^k \mathcal{M}_i}(E) &= \min_{Y \subseteq \hat{E}} \text{rk}_{\mathcal{M}}(Y) + \text{rk}_{\mathcal{N}}(\hat{E} \setminus Y) \\ &= \min_{Z \subseteq E} \text{rk}_{\mathcal{M}}([k] \times Z) + \text{rk}_{\mathcal{N}}([k] \times (E \setminus Z)) \\ &= \min_{Z \subseteq E} \sum_{i=1}^k \text{rk}_{\mathcal{M}_i}(Z) + |E \setminus Z| \end{aligned}$$

as desired.  $\square$

We just showed that Matroid Intersection 1.0.2 implies Matroid Union 2.0.2. Eventually, we will show that they are equivalent. The first step is the following lemma:

**Lemma 2.0.5.** *Let  $\mathcal{M}$  and  $\mathcal{N}$  be matroids on  $E$  and let  $k$  be the maximum size of a common independent set. Then  $\text{rk}_{\mathcal{M} \vee \mathcal{N}^*}(E) = \text{rk}_{\mathcal{N}^*}(E) + k$ .*

*Proof.* Let  $I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$  with  $|I| = k$ . Extend  $I$  to a basis of  $\mathcal{N}$ . Then  $B^c := E \setminus B$  is a basis of  $\mathcal{N}^*$ , so  $I \sqcup B^c$  is independent in  $\mathcal{M} \vee \mathcal{N}^*$ . Hence

$$\text{rk}_{\mathcal{M} \vee \mathcal{N}^*}(E) \geq |I \sqcup B^c| = k + \text{rk}_{\mathcal{N}^*}(E)$$

To show the other direction, choose a basis  $I$  of  $\mathcal{M} \vee \mathcal{N}^*$ , so we may decompose  $I$  as  $I_1 \sqcup I_2$  where  $I_1 \in \mathcal{I}(\mathcal{M})$  and  $I_2 \in \mathcal{I}(\mathcal{N}^*)$ . Among all the feasible  $I_1$  and  $I_2$  subject to this, assume that  $I_1, I_2$  are chosen such that  $|I_2|$  is maximum. Suppose for the contradiction that  $I_2$  is not a basis of  $\mathcal{N}^*$ , then, we may choose  $x \in E \setminus I_2$  such that  $I'_2 := I_2 + x \in \mathcal{I}(\mathcal{N}^*)$ . Now,  $I'_1 := I_1 \setminus \{x\}$  and  $I'_2$  contradict our choice of  $I_1$  and  $I_2$ . Therefore,  $B := E \setminus I_2$  is a basis of  $\mathcal{N}$ . Note that  $I_1 \subseteq B$  and hence,  $I_1 \in \mathcal{I}(\mathcal{N})$ . Together with our assumption  $I_1 \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})$  so  $|I_1| \leq k$ . Therefore

$$\text{rk}_{\mathcal{M} \vee \mathcal{N}^*}(E) = |I_1| + |I_2| \leq k + \text{rk}_{\mathcal{N}^*}(E)$$

as desired.  $\square$

The above lemma gives a key observation of the connection between the "intersection" of  $\mathcal{M}$  and  $\mathcal{N}$ , and the "union" of  $\mathcal{M}$  and  $\mathcal{N}^*$ . Next, we will use this to establish the equivalence of Matroid Intersection and Matroid Union. First, recall the rank function formula of the dual matroids, namely:

$$\text{rk}_{\mathcal{N}^*}(X) = \text{rk}_{\mathcal{N}}(E \setminus X) + |X| - \text{rk}_{\mathcal{N}}(E) \quad (3)$$

and

$$\text{rk}_{\mathcal{N}^*}(X) = \text{rk}_{\mathcal{N}}(E \setminus X) + \text{rk}_{\mathcal{N}^*}(E) - |E \setminus X| \quad (4)$$

*Remark.* To remember the above formulas, think about the following diagram:

$$\begin{array}{c|c}
 X & E \setminus X \\
 \hline
 B & \text{rk}_{\mathcal{N}}(E \setminus X) \\
 \hline
 \overline{B} & \text{rk}_{\mathcal{N}^*}(X)
 \end{array}$$

**Theorem 2.0.6.** *The Matroid Intersection 1.0.2 and the Matroid Union 2.0.2 are equivalent.*

*Proof.* Let  $\mathcal{M}$  and  $\mathcal{N}$  be matroids on  $E$ . First we show that the Matroid Union 2.0.2 for  $\mathcal{M} \vee \mathcal{N}^*$  implies the Matroid Intersection 1.0.2 for  $\mathcal{M}$  and  $\mathcal{N}$ . By Lemma 2.0.5 and (4) we have

$$\begin{aligned}
 \max_{I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})} |I| &= \text{rk}_{\mathcal{M} \vee \mathcal{N}^*}(E) - \text{rk}_{\mathcal{N}^*}(E) \\
 &= \min_{X \subseteq E} \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}^*}(X) + |E \setminus X| - \text{rk}_{\mathcal{N}}(E) \\
 &= \min_{X \subseteq E} \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X)
 \end{aligned}$$

Next, we show that the Matroid Intersection 1.0.2 for  $\mathcal{M}$  and  $\mathcal{N}$  implies the Matroid Union 2.0.2 for  $\mathcal{M} \vee \mathcal{N}^*$ . Again, by Lemma 2.0.5 and (4) we have

$$\begin{aligned}
 \text{rk}_{\mathcal{M} \vee \mathcal{N}^*}(E) &= \max_{I \in \mathcal{I}(\mathcal{M}) \cap \mathcal{I}(\mathcal{N})} |I| + \text{rk}_{\mathcal{N}^*}(E) \\
 &= \min_{X \subseteq E} \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}}(E \setminus X) + \text{rk}_{\mathcal{N}^*}(E) \\
 &= \min_{X \subseteq E} \text{rk}_{\mathcal{M}}(X) + \text{rk}_{\mathcal{N}^*}(X) + |E \setminus X|
 \end{aligned}$$

Finally, we use induction to deduce the Matroid Union 2.0.2 for arbitrarily many matroids. The base case  $k = 2$  is given; for  $k \geq 3$  let  $\mathcal{M}_1, \dots, \mathcal{M}_k$  be matroids on  $E$  with rank functions  $\text{rk}_1, \dots, \text{rk}_k$ . Then

$$\begin{aligned}
 \text{rk}_{\bigvee_{i=1}^k \mathcal{M}_i}(E) &= \text{rk}_{(\bigvee_{i=1}^{k-1} \mathcal{M}_i) \vee \mathcal{M}_k}(E) \\
 &= \min_{X \subseteq E} \text{rk}_{\bigvee_{i=1}^{k-1} \mathcal{M}_i}(X) + \text{rk}_k(X) + |E \setminus X| \\
 &= \min_{X \subseteq E} \min_{Y \subseteq X} \sum_{i=1}^{k-1} \text{rk}_i(Y) + |X \setminus Y| + \text{rk}_k(X) + |E \setminus X| \\
 &= \min_{Y \subseteq X \subseteq E} \sum_{i=1}^{k-1} \text{rk}_i(Y) + \text{rk}_k(X) + |E \setminus Y| \\
 &= \min_{Y \subseteq E} \sum_{i=1}^{k-1} \text{rk}_i(Y) + \text{rk}_k(Y) + |E \setminus Y| \\
 &= \min_{Y \subseteq E} \sum_{i=1}^k \text{rk}_i(Y) + |E \setminus Y|
 \end{aligned}$$

as desired. □